

ON EXTRA PAPER

3. Given the function $y^2 - x^2y = 6, y > 0$ find $\frac{d^2y}{dx^2}$. Calculate the value of $\frac{d^2y}{dx^2}$ at point $P(0, \sqrt{6})$.

(3/14) $y^2 - x^2y = 6$

$$\frac{d}{dx}(y^2) - \frac{d}{dx}(x^2y) = \frac{d}{dx}(6)$$

$$2y \cdot \frac{dy}{dx} - (x^2 \frac{dy}{dx} + 2xy) = 0$$

$$\frac{dy}{dx}(1 - x^2) - 2xy = 0$$

$$\frac{dy}{dx} = \frac{2xy}{1 - x^2}$$

ON PAPER

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{2xy}{1 - x^2} \right)$$

$$= \frac{(1 - x^2) \frac{d}{dx}(2xy) - (2xy) \frac{d}{dx}(1 - x^2)}{(1 - x^2)^2}$$

$$= \frac{(1 - x^2)(2x \frac{dy}{dx} + 2y) - (2xy)(-2x)}{(1 - x^2)^2}$$

$$= \frac{4x^2y + 2y(1 - x^2) + 4x^2y}{(1 - x^2)^2} = \frac{8x^2y + 2y - 2x^2y + 4x^2y}{(1 - x^2)^2} = \frac{6x^2y + 2y}{(1 - x^2)^2}$$

4. Calculate the derivative of the inverse of the following function at corresponding value of $x=3$.

$$f(x) = e^{3x}(3x+1)^2$$

(3, 0) original
(0, 3) inverse

$$f(3) = e^{9}(10) = 100e^9$$

$$\frac{1}{3} (f^{-1})'(3) = \frac{1}{f'(f^{-1}(3))} = \frac{1}{f'(0)}$$

$$(f^{-1})'(100e^9) = \frac{1}{3125475129}$$

$$f'(x) = (3 + \frac{6}{3x+1}) \cdot f(x)$$

Applications (10 marks) $3 \times \frac{5}{7} (100e^9)$

$$y = e^a(a+1)^2$$

$$y = 100e^9$$

$$y = e^{3x}(3x+1)^2$$

$$\ln(y) = \ln(e^{3x}(3x+1)^2)$$

$$= \ln(e^{3x}) + 2\ln(3x+1)$$

$$= 3x$$

1. Find equation for the line tangent and the line normal to the curve $y = \tan^{-1}(2x)$ at $x = -1/2$.

$$y' = \frac{d}{dx}(\tan^{-1}(2x))$$

$$= \frac{1}{1 + (2x)^2} \cdot \frac{d}{dx}(2x)$$

$$1/3 y' = \frac{2}{1 + 4x^2} \text{ at } x = -1/2$$

$$= \frac{2}{1 + 4(-1/2)^2}$$

$$= \frac{2}{1 + 1} = 1$$

$$= \frac{2}{2} = 1$$

$$y = 1(x + \frac{1}{2}) + (\frac{\pi}{4}) \text{ tangent}$$

$$y = -(x + \frac{1}{2}) + (\frac{-\pi}{4}) \text{ Normal}$$

✓

Slope is $\perp$ for points $(-\frac{1}{2}, -\frac{\pi}{4})$

2.5

Deriv of $f(x) = \frac{dy}{dx}$
 $f(x) \cdot (3 + \frac{6}{3x+1})$

take deriv

$$\frac{1}{y} \cdot \frac{dy}{dx} = 3 + 2 \cdot \frac{1}{3x+1} \cdot \frac{3}{1}$$

$$= 3 + \frac{2 \cdot 3}{3x+1}$$

$$= (3 + \frac{6}{3x+1})$$

derivative of a product

2. Given the parametric equations: $x = 2 \sin(1 + 3t)$ and $y = 2t^3$

1/3 a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\frac{dy}{dt} = \frac{d(2t^3)}{dt} = 6t^2$$

$$\frac{dx}{dt} = \frac{d(2 \sin(1+3t))}{dt} = 6 \cos(1+3t)$$

b) find the equation of the tangent at $t = -\frac{1}{3}$

$$\left. \frac{dy}{dx} \right|_{t=-\frac{1}{3}} = \frac{\left(-\frac{1}{3}\right)^2}{\cos(1-1)}$$

$$y \text{ value is } 2 \left(-\frac{1}{3}\right)^3$$

$$2 \cdot \frac{-1}{27}$$

$$y = \frac{1}{9}x + \frac{2}{27}$$

$$x \text{ value is } 2 \sin(1-1)$$

$$2 \sin(0) = 0$$

3. The spread of a virus in a certain school is given by the equation: $P(t) = \frac{180}{1+e^{4-t}}$ where t is the number of days since virus first appeared, and $P(t)$ is the total number of students who have caught the virus to date.

a) What was the initial number of students infected with the virus?

$$1/1 \quad \frac{180}{1+e^{4-0}} = \frac{180}{1+e^4} = 3.23 \text{ so } 3 \text{ students}$$

b) Determine the rate at which the virus spreads.

$$1/2 \quad P'(t) = \frac{d}{dt} \left(\frac{180}{1+e^{4-t}} \right) = 180 \cdot \frac{d}{dt} \left(\frac{1}{1+e^{4-t}} \right) = \frac{180e^{4-t}}{(1+e^{4-t})^2} = P'(t)$$

c) How fast is the virus spreading after 4 days?

$$1/1 \quad P'(4) = \frac{180}{(1+1)^2} = 45 \text{ students/day}$$

Communication (4 marks)

1. Use the information given in the table with x -values of 1, 3, 5, and 7 to answer the following questions:

$$g(3) = 7, g'(3) = 1$$

$$a) \frac{d}{dx} [f \circ g(x)] \text{ at } x = 3$$

$$f'(g(x)) \cdot g'(x) = f'(7) = 1$$

$$b) \frac{d}{dx} \left[\frac{g(x)}{f(x)} \right] \text{ at } x = 1$$

$$f(x) \cdot g'(x) - g(x) \cdot f'(x)$$

$$1/4 \quad \frac{f(x)^2}{(5) \cdot (5) - (3) \cdot (3)}$$

$$\frac{25-9}{25} = \frac{16}{25}$$

$$25$$

$$16 \cdot \frac{dy}{dx} = \frac{dy}{dx} \left[\frac{g(x)}{f(x)} \right]$$

$$c) \frac{d}{dx} [f(x)g(x)] \text{ at } x = 7$$

$$\frac{d}{dx} = f(x)g'(x) + g(x)f'(x) = (3)(7) + (5)(1) = 26$$

$$d) \frac{d}{dx} [f^{-1}(x)] \text{ at } x = 5$$

$$\frac{d}{dx} [f^{-1}(x)] = \frac{1}{f'(1)}$$

$$g'(1) = 3, \frac{d}{dx} [f^{-1}(x)] = \frac{1}{3}$$

$$f'(1) = 3, \frac{d}{dx} [f^{-1}(x)] = \frac{1}{3}$$

x	1	3	5	7
f(x)	5	1	7	3
g(x)	3	7	1	5
f'(x)	3	5	7	1
g'(x)	5	1	3	7

Thinking

1. Use the technique of logarithmic differentiation to find $\frac{dy}{dx}$ if:

$$y = \frac{(x+2)^5(2x-3)^4}{(x+17)^2}$$

$$y = \frac{(x+2)^5(2x-3)^4}{(x+17)^2}$$

$\ln$

$$\ln(y) = \ln\left(\frac{(x+2)^5(2x-3)^4}{(x+17)^2}\right)$$

$\frac{d}{dx}$

$$\frac{d}{dx}(\ln(y)) = \frac{d}{dx}[\ln(x+2) + 4\ln(2x-3) - 2\ln(x+17)]$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 5\left(\frac{1}{x+2}\right) + \frac{8}{2x-3} - \frac{2}{x+17}$$

$$\frac{dy}{dx} = \left(\frac{5}{x+2} + \frac{8}{2x-3} - \frac{2}{x+17}\right)y$$

plug y in

Final derivative

$$\frac{dy}{dx} = \left(\frac{5}{x+2} + \frac{8}{2x-3} - \frac{2}{x+17}\right) \frac{(x+2)^5(2x-3)^4}{(x+17)^2}$$

2. Given the curve $x^2 - xy + y^2 = 9$, determine the point(s) where the tangents are vertical.

$\frac{d}{dx}$

$$\frac{d}{dx}(x^2 - xy + y^2) = 0$$

$$2x - x\frac{dy}{dx} - y + 2y\frac{dy}{dx} = 0$$

$$14\frac{dy}{dx}(2y-x) = y-2x$$

$$\frac{dy}{dx} = \frac{y-2x}{2y-x}$$

← set bottom to zero to have tangent

$$2y-x=0 \quad 2y=x \quad \leftarrow \text{plug back in}$$

Points

$$(2y)^2 - (2y)(y) + y^2 = 9$$

$$4y^2 - 2y^2 + y^2 = 9$$

$$3y^2$$

$$y = \pm\sqrt{3} \quad x = \pm 2\sqrt{3}$$

$$P_1 (\sqrt{3}, 2\sqrt{3})$$

$$P_2 (-\sqrt{3}, -2\sqrt{3})$$

1. Differentiate the following.

a) $y = \sqrt{1-x^2} - \sin^{-1}x$, state domain of f'

$$\begin{aligned} y' &= \frac{1}{2\sqrt{1-x^2}} \cdot (-2x) - \frac{1}{\sqrt{1-x^2}} \\ &= \frac{-x}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} \\ &= \frac{-x-1}{\sqrt{1-x^2}} \end{aligned}$$

f' domain = $-1 < x < 1$ not =

c) $y = \ln(e^{-x} + xe^{-x})$, state the domain of f

$$f'_{\text{domain}} = x > -1 \quad \checkmark \quad \ln(e^{-x}(1+x))$$

$$y' = \frac{1}{e^{-x} + xe^{-x}} \cdot \frac{d}{dx}(e^{-x} + xe^{-x})$$

$$y' = \frac{1}{e^{-x} + xe^{-x}} (-e^{-x} + -xe^{-x} + e^{-x})$$

$$y' = \frac{-xe^{-x}}{e^{-x} + xe^{-x}} = y' = \frac{-x}{1+x} \quad \checkmark$$

2. Use implicit differentiation to find the derivative of the following:

$$e^y \cdot \cos x = 1 + \sin(xy)$$

$$\frac{d}{dx}(e^y \cdot \cos x) = \frac{d}{dx}(1 + \sin(xy))$$

$$e^y \frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}e^y = -\cos(xy) \cdot \frac{d}{dx}(xy)$$

$$-\sin(x) \cdot e^y + e^y \cos(xy) \frac{d}{dx} = -\cos(xy) (x \frac{dy}{dx} + y)$$

$$-\sin(x) e^y - \cos(xy) y = \cos(xy) (x) \frac{dy}{dx} - e^y \cos(x) \frac{dy}{dx}$$

$$-\sin(x) e^y - \cos(xy) y = \frac{dy}{dx} (\cos(xy)(x) - e^y \cos(x))$$

$$\frac{-\sin(x) e^y - \cos(xy) y}{\cos(xy)(x) - e^y \cos(x)} = \frac{dy}{dx} \quad \checkmark$$

b) $y = \log\left[\frac{x+1}{x-2}\right]$

$$y = \log(x+1) - \frac{1}{2} \log(x-2)$$

$$y' = \frac{1}{\ln 10 (x+1)} - \frac{1}{2} \left(\frac{1}{\ln 10 (x-2)} \right)$$

$$y' = \left[\frac{1}{x+1} - \frac{1}{2(x-2)} \right] \frac{1}{\ln 10}$$

$$y' = \frac{2x-2-(x+1)}{2(x+1)(x-2)} = \frac{x-3}{2(x+1)(x-2)}$$

d) $y = \frac{x^2-1}{5x^2+1}$

find $\frac{dy}{dx}$ at $x=1$

$$\ln(y) = \ln\left(\frac{x^2-1}{5x^2+1}\right)$$

$$\frac{1}{y} \left(\frac{dy}{dx} \right) = \frac{1}{\ln\left(\frac{x^2-1}{5x^2+1}\right)} \ln(5)$$

$$\left(\frac{1}{y} \right) \frac{dy}{dx} = \ln(5) \cdot \frac{(x^2+1)^2}{(x^2+1)^2 - (x^2-1)^2}$$

$$\frac{dy}{dx} = \ln(5) \cdot \frac{2x(x^2+1)}{(x^2+1)^2 - (x^2-1)^2}$$

$$\frac{dy}{dx} = \frac{4x}{(x^2+1)^2} \cdot 5 \ln(5)$$

$x=1$
 $y=1$

$$\frac{d}{dx}(e^y \cdot \cos x) = \frac{d}{dx}(1 + \sin(xy))$$

$$e^y \frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}e^y = -\cos(xy) \cdot \frac{d}{dx}(xy)$$

$$-\sin(x) \cdot e^y + e^y \cos(xy) \frac{d}{dx} = -\cos(xy) (x \frac{dy}{dx} + y)$$

$$-\sin(x) e^y - \cos(xy) y = \cos(xy) (x) \frac{dy}{dx} - e^y \cos(x) \frac{dy}{dx}$$

$$-\sin(x) e^y - \cos(xy) y = \frac{dy}{dx} (\cos(xy)(x) - e^y \cos(x))$$

$$\frac{-\sin(x) e^y - \cos(xy) y}{\cos(xy)(x) - e^y \cos(x)} = \frac{dy}{dx} \quad \checkmark$$

(-2)